VAT Manual

Versatile Alignment Tool

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Description

VAT (Versatile Alignment Tool) is a high-performance sequence alignment framework for DNA and protein sequences. It supports multiple alignment paradigms including: 1) whole-genome read mapping; 2) RNA-seq splice alignment; 3) metagenomic alignment; 4) circular RNA alignment; 5) whole genome alignment 5) DNA homology search; 6) protein homology; 7) translated DNA–protein alignment. VAT is designed to provide a unified alignment framework with high performance and flexible seeding strategies.

Command Overview

General Syntax

The general command-line syntax of VAT is:

VAT <command> [options]

Commands

VAT provides several commands for database construction, sequence alignment, and result processing.

makevatdb

- Build a VAT database from a reference FASTA file.

dna

- Perform DNA-to-DNA sequence alignment.

protein

- Perform protein-to-protein sequence alignment.

blastx

- Perform translated DNA-to-protein alignment.

view

- View or convert VAT alignment archives.

dna2pro

- Translate DNA sequences into protein sequences.

Getting Started

Basic workflow

The basic VAT workflow consists of two main steps.

1. Building a VAT database

In the first step, the user provides reference sequences in FASTA format. VAT uses these sequences to build a searchable database that will be used during sequence alignment. The database construction step is performed using the makevatdb command. The generated database files are stored on disk and only need to be created once for each reference dataset. This step prepares the reference sequences for efficient indexing and fast alignment during subsequent searches.

2. Running sequence alignment

In the second step, the user performs sequence alignment against the VAT database generated in the first step. Depending on the analysis task, different VAT commands can be used:

- **dna** – DNA-to-DNA sequence alignment
- **protein** – protein-to-protein homology search
- **blastx** – translated DNA-to-protein alignment

The input sequences can be provided in FASTA or FASTQ format. VAT processes these sequences and produces alignment results that can be inspected or converted using the view command. The alignment process can be controlled through various command-line parameters, allowing users to adjust sensitivity, performance, and output format according to their analysis needs. The general VAT command-line format is:

```
VAT <command> [options]
```

If an option accepts multiple values, they are typically separated by spaces. In some cases, comma-separated values may also be supported. This description matches the purpose of VAT (Versatile Alignment Tool)—a multi-purpose sequence aligner that supports nucleotide alignment, protein homology search, and translated searches within a unified framework.

Generating VAT Database

Command

```
VAT <command>
```

Synopsis

```
VAT <command> [options]
```

Examples

```
VAT makevatdb --in [reference.fa] --dbtype [Database types] --threads [num of threads]
```

```
VAT makevatdb --in reference.fa --dbtype nucl --threads 8
```

```
VAT makevatdb --in Pfam.fa --dbtype prot --threads 8
```

Description

Build a VAT searchable database from reference sequences in FASTA format. The generated database is used for subsequent VAT alignment commands.

Parameters

--in

Type: FASTA file
Default: none

Input reference FASTA file.

--dbtype

Type: nucl | prot
Default: nucl

Specifies the type of reference database.

- nucl – nucleotide database
- prot – protein database

--threads

Type: integer
Default: 4

Number of CPU threads used during database construction.

Output

VAT generates database files stored in the internal .vatf format, which is optimized for fast sequence alignment.

Running Alignment

VAT provides several alignment modes designed for different biological applications, including whole-genome alignment, RNA-seq analysis, metagenomic search, and protein homology detection. These modes can be selected using the corresponding command-line options.

Alignment Modes

--wgs

Whole-genome sequencing alignment mode.

Optimized for aligning sequencing reads to a reference genome.

--wga

Whole-genome alignment mode.

Used for aligning large genomic sequences between species or assemblies.

--splice

Splice-aware alignment mode for RNA-seq data.

Supports detection of exon–exon junctions.

--circ

Circular RNA detection mode.

Identifies potential circular RNA junctions.

--chimera

Chimeric alignment detection mode.

Detects reads originating from structural rearrangements or fusion events.

--metagenomic

Metagenomic alignment mode.

Designed for mapping reads to large collections of microbial reference sequences.

--dnah

DNA homology search mode.

Performs similarity searches between DNA sequences.

--long

Long-read alignment mode.

Optimized for long sequencing reads produced by platforms such as PacBio or Oxford Nanopore.

--SEN

Sensitive protein alignment mode.

Provides increased sensitivity for detecting distant protein homologs.

Basic options

The following parameters control the core behavior of VAT alignment.

-p, --threads

Type: integer
Default: 4

Number of CPU threads used during alignment.

-S, --seed_len

Type: integer
Default: 15

Length of the seed (k-mer) used for initial sequence matching.
Smaller values increase sensitivity but may reduce alignment speed.

-d, --db

Type: file

Path to the VAT reference database file.

-q, --query

Type: file

Input query sequence file.
Supported formats typically include FASTA and FASTQ.

--gapopen

Type: integer

Gap opening penalty applied when introducing a new gap during alignment.

--gapextend

Type: integer

Gap extension penalty applied when extending an existing gap.

--match

Type: integer

Score assigned to a matching base.

--mismatch

Type: integer

Penalty applied to mismatched base.

-o, --out

Type: file

Output file used to store alignment results.

-a, --vaa

Type: file (optional)

Archive file used to store VAT alignment results in the internal .vatr format for the whole genome alignment part. This archive can later be inspected or converted using the vat view command.

-f, --outfmt

Type: tab | sam | paf | pairwise
Default: tab

Specifies the output format for alignment results.

- tab – tabular alignment output
- sam – SAM format for compatibility with downstream tools
- paf – Pairwise mApping Format
- pairwise – human-readable pairwise alignment format

Example

```
VAT dna --db reference.fa --query reads.fq -f tab -p 8 -o
results.tab
```

Whole Genome Sequencing Alignment

VAT provides a high-performance alignment mode designed for whole genome sequencing (WGS) data. This mode supports both short-read and long-read sequencing technologies and is optimized for fast and accurate genome-scale alignment. The WGS mode can be enabled using the `--wgs` option in the VAT dna command.

Usage

Short-read alignment:

```
VAT dna -d reference.fa -q reads.fa --wgs -o output.sam -f sam -p
8
```

Long-read alignment:

```
VAT dna -d reference.fa -q long_reads.fa --wgs --long -o
output.sam -f sam
```

Custom seed length:

```
VAT dna -d reference.fa -q reads.fa --wgs -S 15 -o output.sam -f
sam
```

Parameters

--wgs

Enable whole genome sequencing alignment mode.
This mode is optimized for mapping sequencing reads to a reference genome.

--long

Enable long-read alignment mode.
Designed for sequencing platforms that generate long reads (e.g., PacBio or Oxford Nanopore).
If not specified, VAT assumes short-read alignment.

-S, --seed_len

Type: integer
Default: 15

Seed length (k-mer size) used for initial sequence matching.
Smaller seed sizes increase alignment sensitivity but may reduce speed.

Output Formats

VAT supports multiple output formats for alignment results. The output format can be specified using the -f or --outfmt option.

SAM Format (-f sam)

The SAM (Sequence Alignment/Map) format is widely used for representing sequence alignments and is compatible with many downstream analysis tools.

SAM Header

Header lines begin with @ and describe metadata about the alignment file.

Typical header records include:

@HD – Header line
Must include SO:unsorted to indicate unsorted SAM output.

Example:

@HD VN:1.5 SO:unsorted

@SQ – Reference sequence information
One line is provided for each reference sequence.

Fields include:

- SN – reference sequence name
- LN – reference sequence length

Example:

@SQ SN:chr1 LN:1000

@PG – Program information

Contains metadata about the program that generated the SAM file.

Example:

@PG ID:VAT PN:VAT VN:1.0

SAM Alignment Fields

Each alignment line contains 11 mandatory columns followed by optional tags.

- **QNAME** – Query (read) name
- **FLAG** – Bitwise alignment flag indicating strand and mapping status
- **RNAME** – Reference sequence name
- **POS** – 1-based leftmost mapping position on the reference
- **MAPQ** – Mapping quality score
- **CIGAR** – CIGAR string describing alignment operations
- **RNEXT** – Reference name of the next read in the template (* for single-end reads)
- **PNEXT** – Position of the next read in the template (0 for single-end reads)

- **TLEN** – Template length (0 for single-end reads)
- **SEQ** – Query sequence
- **QUAL** – Base quality string (* if not available)

For simple alignments, VAT typically sets the FLAG value to:

- 0 – forward strand alignment
- 16 – reverse strand alignment

Optional SAM Tags

VAT may include additional tags to describe alignment properties.

Tag	Type	Description
NM:i	integer	Number of mismatches
AS:i	integer	Alignment score
id:f	float	Percent identity (0–100, two decimal places)
cg:Z	string	CIGAR string (duplicate representation for convenience)

PAF Format (-f paf)

VAT also supports output in PAF (Pairwise mApping Format), a lightweight format widely used for long-read alignment tools.

Each line represents one alignment.

- **Qseqid** – Query sequence name
- **query_length** – Full length of the query sequence
- **query_start** – Alignment start position on the query
- **query_end** – Alignment end position on the query
- **strand** – Alignment strand (+ or -)
- **Sseqid** – Target (reference) sequence name
- **target_length** – Full length of the target sequence
- **target_start** – Alignment start position on the target
- **target_end** – Alignment end position on the target
- **match_bases** – Number of matching bases in the alignment
- **MAPQ** – Mapping quality score (0–60)

RNA-seq Splice Alignment

VAT provides a high-performance alignment mode designed for RNA-seq splice data. This mode supports both short-read and long-read sequencing technologies and is optimized for fast and accurate genome-scale alignment. The RNA-seq splice mode can be enabled using the --splice option in the VAT dna command.

Usage

Short-read alignment:

```
VAT dna -d reference.fa -q reads.fa --splice -o output.sam -f sam
-p 8
```

Long-read alignment:

```
VAT dna -d reference.fa -q long_reads.fa --splice --long -o
output.sam -f sam
```

Custom seed length:

```
VAT dna -d reference.fa -q reads.fa --splice -S 15 -o output.sam
-f sam
```

Parameters

--splice

Enable RNA-seq splice-aware alignment mode.
This mode is optimized for mapping sequencing reads to a reference genome.

--long

Enable long-read alignment mode.
Designed for sequencing platforms that generate long reads (e.g., PacBio or Oxford Nanopore).
If not specified, VAT assumes short-read alignment.

-S, --seed_len

Type: integer
Default: 15

Seed length (k-mer size) used for initial sequence matching.
Smaller seed sizes increase alignment sensitivity but may reduce speed.

Output Formats

VAT supports multiple output formats for alignment results. The output format can be specified using the -f or --outfmt option.

SAM Format (-f sam)

The SAM (Sequence Alignment/Map) format is widely used for representing sequence alignments and is compatible with many downstream analysis tools.

SAM Header

Header lines begin with @ and describe metadata about the alignment file.

Typical header records include:

@HD – Header line
Must include SO:unsorted to indicate unsorted SAM output.

Example:

@HD VN:1.5 SO:unsorted

@SQ – Reference sequence information
One line is provided for each reference sequence.

Fields include:

- SN – reference sequence name
- LN – reference sequence length

Example:

@SQ SN:chr1 LN:1000

@PG – Program information

Contains metadata about the program that generated the SAM file.

Example:

@PG ID:VAT PN:VAT VN:1.0

SAM Alignment Fields

Each alignment line contains 11 mandatory columns followed by optional tags.

- **QNAME** – Query (read) name
- **FLAG** – Bitwise alignment flag indicating strand and mapping status
- **RNAME** – Reference sequence name
- **POS** – 1-based leftmost mapping position on the reference
- **MAPQ** – Mapping quality score
- **CIGAR** – CIGAR string describing alignment operations
- **RNEXT** – Reference name of the next read in the template (* for single-end reads)
- **PNEXT** – Position of the next read in the template (0 for single-end reads)

- **TLEN** – Template length (0 for single-end reads)
- **SEQ** – Query sequence
- **QUAL** – Base quality string (* if not available)

For simple alignments, VAT typically sets the FLAG value to:

- 0 – forward strand alignment
- 16 – reverse strand alignment

Optional SAM Tags

VAT may include additional tags to describe alignment properties.

Tag	Type	Description
NM:i	integer	Number of mismatches
AS:i	integer	Alignment score
id:f	float	Percent identity (0–100, two decimal places)
cg:Z	string	CIGAR string (duplicate representation for convenience)

PAF Format (-f paf)

VAT also supports output in PAF (Pairwise mApping Format), a lightweight format widely used for long-read alignment tools.

Each line represents one alignment.

- **Qseqid** – Query sequence name
- **query_length** – Full length of the query sequence
- **query_start** – Alignment start position on the query
- **query_end** – Alignment end position on the query
- **strand** – Alignment strand (+ or -)
- **Sseqid** – Target (reference) sequence name
- **target_length** – Full length of the target sequence
- **target_start** – Alignment start position on the target
- **target_end** – Alignment end position on the target
- **match_bases** – Number of matching bases in the alignment
- **MAPQ** – Mapping quality score (0–60)

Circular RNA Alignment

VAT provides a high-performance alignment mode designed for Circular RNA data. This mode supports both short-read and long-read sequencing technologies and is optimized for fast and accurate genome-scale alignment. The Circular RNA Alignment mode can be enabled using the --circ option in the VAT dna command.

Usage

```
VAT dna -d reference.fa -q reads.fa --circ -o output.sam -f sam -p 8
```

```
VAT dna -d reference.fa -q reads.fa --circ -S 15 -o output.sam -f sam
```

Parameters

--circ

Enable circular RNA alignment mode.
This mode is optimized for mapping sequencing reads to a reference genome.

-S, --seed_len

Type: integer
Default: 15

Seed length (k-mer size) used for initial sequence matching.
Smaller seed sizes increase alignment sensitivity but may reduce speed.

Output Formats

VAT supports multiple output formats for alignment results. The output format can be specified using the -f or --outfmt option.

SAM Format (-f sam)

The SAM (Sequence Alignment/Map) format is widely used for representing sequence alignments and is compatible with many downstream analysis tools.

SAM Header

Header lines begin with @ and describe metadata about the alignment file.

Typical header records include:

@HD – Header line
Must include SO:unsorted to indicate unsorted SAM output.

Example:

@HD VN:1.5 SO:unsorted

@SQ – Reference sequence information
One line is provided for each reference sequence.

Fields include:

- SN – reference sequence name
- LN – reference sequence length

Example:

@SQ SN:chr1 LN:1000

@PG – Program information

Contains metadata about the program that generated the SAM file.

Example:

@PG ID:VAT PN:VAT VN:1.0

SAM Alignment Fields

Each alignment line contains 11 mandatory columns followed by optional tags.

- **QNAME** – Query (read) name
- **FLAG** – Bitwise alignment flag indicating strand and mapping status
- **RNAME** – Reference sequence name
- **POS** – 1-based leftmost mapping position on the reference
- **MAPQ** – Mapping quality score
- **CIGAR** – CIGAR string describing alignment operations
- **RNEXT** – Reference name of the next read in the template (* for single-end reads)
- **PNEXT** – Position of the next read in the template (0 for single-end reads)
- **TLEN** – Template length (0 for single-end reads)
- **SEQ** – Query sequence
- **QUAL** – Base quality string (* if not available)

For simple alignments, VAT typically sets the FLAG value to:

- 0 – forward strand alignment
- 16 – reverse strand alignment

Tabular Format (-f tab)

VAT also supports a BLAST-like tabular output format, which is compact and easy to parse for downstream analysis.

Tabular Output Fields

- **Qseqid** – Query sequence identifier (read name).
- **Sseqid** – Subject sequence identifier (reference or database sequence name).
- **Pident** – Percentage identity within the aligned region, ranging from 0 to 100.
- **Length** – Total alignment length, including matches, mismatches, and gaps.
- **Mismatch** - Number of mismatching bases or amino acids in the alignment.

- **Gapopen** – Number of gap openings (insertion or deletion segments).
- **Qstart** – Alignment start position on the query sequence (1-based coordinate).
- **Qend** – Alignment end position on the query sequence (1-based coordinate).
- **Sstart** – Alignment start position on the subject (reference) sequence (1-based coordinate).
- **Send** – Alignment end position on the subject sequence (1-based coordinate).
- **Evalue** – Expectation value (E-value) of the alignment. Smaller values indicate more significant matches.
- **Bitscore** – Bit score of the alignment. Larger values indicate higher alignment quality.

Metagenomic Alignment

VAT provides a high-performance alignment mode designed for metagenomic data. This mode supports both short-read and long-read sequencing technologies and is optimized for fast and accurate genome-scale alignment. The metagenomic mode can be enabled using the `--metagenomic` option in the `VAT dna` command.

Usage

```
VAT dna -d reference.fa -q reads.fa --metagenomic -o output.sam -f sam -p 8
```

```
VAT dna -d reference.fa -q reads.fa --metagenomic -S 15 -o output.sam -f sam
```

Parameters

--metagenomic

Enable metagenomic alignment mode.
This mode is optimized for mapping sequencing reads to a reference genome.

--long

Enable long-read alignment mode.
Designed for sequencing platforms that generate long reads (e.g., PacBio or Oxford Nanopore).
If not specified, VAT assumes short-read alignment.

-S, --seed_len

Type: integer
Default: 15

Seed length (k-mer size) used for initial sequence matching.
Smaller seed sizes increase alignment sensitivity but may reduce speed.

Output Formats

VAT supports multiple output formats for alignment results. The output format can be specified using the -f or --outfmt option.

SAM Format (-f sam)

The SAM (Sequence Alignment/Map) format is widely used for representing sequence alignments and is compatible with many downstream analysis tools.

SAM Header

Header lines begin with @ and describe metadata about the alignment file.

Typical header records include:

@HD – Header line
Must include SO:unsorted to indicate unsorted SAM output.

Example:

@HD VN:1.5 SO:unsorted

@SQ – Reference sequence information
One line is provided for each reference sequence.

Fields include:

- SN – reference sequence name
- LN – reference sequence length

Example:

@SQ SN:chr1 LN:1000

@PG – Program information

Contains metadata about the program that generated the SAM file.

Example:

@PG ID:VAT PN:VAT VN:1.0

SAM Alignment Fields

Each alignment line contains 11 mandatory columns followed by optional tags.

- **QNAME** – Query (read) name
- **FLAG** – Bitwise alignment flag indicating strand and mapping status
- **RNAME** – Reference sequence name
- **POS** – 1-based leftmost mapping position on the reference
- **MAPQ** – Mapping quality score
- **CIGAR** – CIGAR string describing alignment operations
- **RNEXT** – Reference name of the next read in the template (* for single-end reads)
- **PNEXT** – Position of the next read in the template (0 for single-end reads)
- **TLEN** – Template length (0 for single-end reads)
- **SEQ** – Query sequence
- **QUAL** – Base quality string (* if not available)

For simple alignments, VAT typically sets the FLAG value to:

- 0 – forward strand alignment
- 16 – reverse strand alignment

Tabular Format (-f tab)

VAT also supports a BLAST-like tabular output format, which is compact and easy to parse for downstream analysis.

Tabular Output Fields

- **Qseqid** – Query sequence identifier (read name).
- **Sseqid** – Subject sequence identifier (reference or database sequence name).
- **Pident** – Percentage identity within the aligned region, ranging from 0 to 100.
- **Length** – Total alignment length, including matches, mismatches, and gaps.
- **Mismatch** - Number of mismatching bases or amino acids in the alignment.
- **Gapopen** – Number of gap openings (insertion or deletion segments).
- **Qstart** – Alignment start position on the query sequence (1-based coordinate).
- **Qend** – Alignment end position on the query sequence (1-based coordinate).
- **Sstart** – Alignment start position on the subject (reference) sequence (1-based coordinate).
- **Send** – Alignment end position on the subject sequence (1-based coordinate).
- **Evalue** – Expectation value (E-value) of the alignment. Smaller values indicate more significant matches.
- **Bitscore** – Bit score of the alignment. Larger values indicate higher alignment quality.

Whole Genome Alignment

Whole Genome Alignment (WGA) mode enables VAT to perform large-scale genome-to-genome alignment. This mode is designed for comparing complete genomes or long genomic sequences and identifying large homologous regions between the reference and query genomes. VAT first performs DNA sequence alignment using the dna command with the --wga option enabled. The resulting alignment archive can then be converted to a human-readable format using the view command.

Usage

```
VAT dna -d genome.fa -q species.fa --wga -a output -f tab -p 32
VAT view -p 32 -a output.vatr -o output.tab -f tab
```

Parameters

--wga

Whole Genome Alignment mode.
This mode is optimized for mapping genome sequences to a reference genome.

-S, --seed_len

Type: integer
Default: 15

Seed length (k-mer size) used for initial sequence matching.
Smaller seed sizes increase alignment sensitivity but may reduce speed.

Output Formats

VAT generates an alignment archive file in .vatr format, which stores the alignment results in a compressed internal representation. The archive can be converted to other formats using the VAT view command. VAT supports multiple output formats for alignment results. The output format can be specified using the -f or --outfmt option.

```
VAT view -p 32 -a output.vatr -o output.tab -f tab
```

Tabular Format (-f tab)

VAT also supports a BLAST-like tabular output format, which is compact and easy to parse for downstream analysis.

Tabular Output Fields

- **Qseqid** – Query sequence identifier (read name).
- **Sseqid** – Subject sequence identifier (reference or database sequence name).
- **Pident** – Percentage identity within the aligned region, ranging from 0 to 100.
- **Length** – Total alignment length, including matches, mismatches, and gaps.
- **Mismatch** - Number of mismatching bases or amino acids in the alignment.
- **Gapopen** – Number of gap openings (insertion or deletion segments).
- **Qstart** – Alignment start position on the query sequence (1-based coordinate).

- **Qend** – Alignment end position on the query sequence (1-based coordinate).
- **Sstart** – Alignment start position on the subject (reference) sequence (1-based coordinate).
- **Send** – Alignment end position on the subject sequence (1-based coordinate).
- **Evalue** – Expectation value (E-value) of the alignment. Smaller values indicate more significant matches.
- **Bitscore** – Bit score of the alignment. Larger values indicate higher alignment quality.

Pairwise Format (-f pairwise)

VAT also supports a **BLAST-like pairwise output format**, which presents alignments in a human-readable form for manual inspection and debugging.

- **Qseqid** – Query sequence identifier (read name).
- **Sseqid** – Subject sequence identifier (reference or database sequence name), together with the total sequence length.
- **Score** – Alignment score reported as a bit score and the corresponding raw alignment score.
- **Evalue** – Expectation value (E-value) of the alignment. Smaller values indicate more significant matches.
- **Identities** – Number and percentage of identical residues within the aligned region.
- **Positives** – Number and percentage of similar residues (reported for protein alignments).
- **Gaps** – Number and percentage of gap openings in the alignment.
- **Frame** – Translation frame of the query sequence for translated alignments (reported in blastx mode).
- **Query alignment line** – Displays the aligned query sequence segment together with its start and end coordinates (1-based).
- **Match line** – A visual representation of matches and mismatches between the query and subject sequences. Identical residues are typically indicated by symbols such as |.
- **Subject alignment line** – Displays the aligned subject sequence segment together with its start and end coordinates (1-based).
- **Line width** – Alignments are printed in segments of 60 characters per line, following the traditional BLAST pairwise layout.

PAF Format (-f paf)

VAT also supports output in PAF (Pairwise mApping Format), a lightweight format widely used for long-read alignment tools.

Each line represents one alignment.

- **Qseqid** – Query sequence name
- **query_length** – Full length of the query sequence
- **query_start** – Alignment start position on the query

- **query_end** – Alignment end position on the query
- **strand** – Alignment strand (+ or -)
- **Sseqid** – Target (reference) sequence name
- **target_length** – Full length of the target sequence
- **target_start** – Alignment start position on the target
- **target_end** – Alignment end position on the target
- **match_bases** – Number of matching bases in the alignment

DNA Homology

DNA Homology Search mode enables VAT to perform large-scale genome-to-genome alignment. This mode is designed for comparing complete genomes or long genomic sequences and identifying large homologous regions between the reference and query genomes. VAT first performs DNA sequence alignment using the dna command with the --dnah option enabled. The resulting alignment archive can then be converted to a human-readable format using the view command.

Usage

```
VAT dna -d genome.fa -q species.fa --dnah -a output -f tab -p 32
```

```
VAT view -p 32 -a output.vatr -o output.tab -f tab
```

```
VAT dna -d genome.fa -q species.fa --dnah -o output.tab -f tab -p 32
```

Parameters

--dnah

DNA Homology Search mode.
This mode is optimized for mapping genome sequences to a reference genome.

-S, --seed_len

Type: integer
Default: 15

Seed length (k-mer size) used for initial sequence matching.
Smaller seed sizes increase alignment sensitivity but may reduce speed.

Output Formats

VAT generates an alignment archive file in .vatr format, which stores the alignment results in a compressed internal representation. The archive can be converted to other formats using the VAT view command. VAT supports multiple output formats for alignment results. The output format can be specified using the -f or --outfmt option.

```
VAT view -p 32 -a output.vatr -o output.tab -f tab
```

Tabular Format (-f tab)

VAT also supports a BLAST-like tabular output format, which is compact and easy to parse for downstream analysis.

Tabular Output Fields

- **Qseqid** – Query sequence identifier (read name).
- **Sseqid** – Subject sequence identifier (reference or database sequence name).
- **Pident** – Percentage identity within the aligned region, ranging from 0 to 100.
- **Length** – Total alignment length, including matches, mismatches, and gaps.
- **Mismatch** - Number of mismatching bases or amino acids in the alignment.

- **Gapopen** – Number of gap openings (insertion or deletion segments).
- **Qstart** – Alignment start position on the query sequence (1-based coordinate).
- **Qend** – Alignment end position on the query sequence (1-based coordinate).
- **Sstart** – Alignment start position on the subject (reference) sequence (1-based coordinate).
- **Send** – Alignment end position on the subject sequence (1-based coordinate).
- **Evalue** – Expectation value (E-value) of the alignment. Smaller values indicate more significant matches.
- **Bitscore** – Bit score of the alignment. Larger values indicate higher alignment quality.

Protein Alignment

Protein alignment mode enables VAT to perform sequence similarity searches based on protein-level alignment. This mode is designed for comparing protein sequences and identifying homologous proteins between the reference database and the query dataset.

The alignment is performed using the VAT protein command. By aligning sequences at the protein level, VAT can detect conserved functional domains and distant homologs that may not be detectable using direct nucleotide alignment. This approach is particularly useful for protein annotation, functional prediction, and comparative genomics studies.

Usage

```
VAT protein -d Pfam.fa -q proteins.fa -a output -f tab -p 32
VAT view -p 32 -a output.vatr -o output.tab -f tab
VAT protein -d Pfam.fa -q proteins.fa -o output -f tab -p 32
VAT protein -d Pfam.fa -q proteins.fa –SEN -o output -f tab -p 32
```

Parameters

--SEN

Sensitive protein alignment mode.

This mode increases alignment sensitivity and is useful for detecting distant protein homologs. It may increase computational cost compared to the default mode.

-S, --seed_len

Type: integer
Default: 15

Seed length (k-mer size) used for initial sequence matching.
Smaller seed sizes increase alignment sensitivity but may reduce speed.

-N, --num_shapes

Type: integer
Default: 0

Number of seed shapes used during the seeding phase of alignment (protein mode only).

If set to 0, VAT will use all available seed shapes. Increasing the number of seed shapes generally improves alignment sensitivity but may increase computational cost. This option is primarily used in protein alignment mode to control the diversity of seed patterns used for detecting homologous regions.

Output Formats

VAT generates an alignment archive file in .vatr format, which stores the alignment results in a compressed internal representation. The archive can be converted to other formats

using the VAT view command. VAT supports multiple output formats for alignment results. The output format can be specified using the -f or --outfmt option.

```
VAT view -p 32 -a output.vatr -o output.tab -f tab
```

Tabular Format (-f tab)

VAT also supports a BLAST-like tabular output format, which is compact and easy to parse for downstream analysis.

Tabular Output Fields

- **Qseqid** – Query sequence identifier (read name).
- **Sseqid** – Subject sequence identifier (reference or database sequence name).
- **Pident** – Percentage identity within the aligned region, ranging from 0 to 100.
- **Length** – Total alignment length, including matches, mismatches, and gaps.
- **Mismatch** - Number of mismatching bases or amino acids in the alignment.
- **Gapopen** – Number of gap openings (insertion or deletion segments).
- **Qstart** – Alignment start position on the query sequence (1-based coordinate).
- **Qend** – Alignment end position on the query sequence (1-based coordinate).
- **Sstart** – Alignment start position on the subject (reference) sequence (1-based coordinate).
- **Send** – Alignment end position on the subject sequence (1-based coordinate).
- **Evalue** – Expectation value (E-value) of the alignment. Smaller values indicate more significant matches.
- **Bitscore** – Bit score of the alignment. Larger values indicate higher alignment quality.

Pairwise Format (-f pairwise)

VAT also supports a **BLAST-like pairwise output format**, which presents alignments in a human-readable form for manual inspection and debugging.

- **Qseqid** – Query sequence identifier (read name).
- **Sseqid** – Subject sequence identifier (reference or database sequence name), together with the total sequence length.
- **Score** – Alignment score reported as a bit score and the corresponding raw alignment score.
- **Evalue** – Expectation value (E-value) of the alignment. Smaller values indicate more significant matches.
- **Identities** – Number and percentage of identical residues within the aligned region.
- **Positives** – Number and percentage of similar residues (reported for protein alignments).
- **Gaps** – Number and percentage of gap openings in the alignment.
- **Frame** – Translation frame of the query sequence for translated alignments (reported in blastx mode).
- **Query alignment line** – Displays the aligned query sequence segment together with its start and end coordinates (1-based).

- **Match line** – A visual representation of matches and mismatches between the query and subject sequences. Identical residues are typically indicated by symbols such as |.
- **Subject alignment line** – Displays the aligned subject sequence segment together with its start and end coordinates (1-based).
- **Line width** – Alignments are printed in segments of 60 characters per line, following the traditional BLAST pairwise layout.

Translated Alignment

Translated alignment mode enables VAT to perform DNA-to-protein homology searches by translating nucleotide sequences and aligning them against protein reference databases. This mode is useful for detecting coding regions and identifying homologous proteins when working with genomic or transcriptomic sequences.

In this mode, VAT translates nucleotide sequences in multiple reading frames and performs protein-level alignment against the reference database. This approach improves sensitivity for detecting conserved protein domains and distant homologs that may not be detectable through direct nucleotide alignment.

The alignment is performed using the VAT blastx command. The resulting alignment results can be stored in a VAT archive file (.vatr) or written directly to an output file (optional). The archive file can later be converted into human-readable formats using the VAT view command.

Usage

```
VAT blastx -d Pfam.fa -q reads.fa -o output.tab -f tab -p 8
```

```
VAT blastx -d Pfam.fa -q reads.fa -o output.tab -f tab -p 8 --SEN
```

Parameters

--SEN

Sensitive alignment mode for translated searches.
This mode increases sensitivity for detecting distant homologs

-S, --seed_len

Type: integer
Default: 15

Seed length (k-mer size) used for initial sequence matching.
Smaller seed sizes increase alignment sensitivity but may reduce speed.

-N, --num_shapes

Type: integer
Default: 0

Number of seed shapes used during the seeding phase of alignment (protein mode only).

If set to 0, VAT will use all available seed shapes. Increasing the number of seed shapes generally improves alignment sensitivity but may increase computational cost. This option is primarily used in protein alignment mode to control the diversity of seed patterns used for detecting homologous regions.

Output Formats

VAT generates an alignment archive file in .vatr format, which stores the alignment results in a compressed internal representation. The archive can be converted to other formats

using the VAT view command. VAT supports multiple output formats for alignment results. The output format can be specified using the -f or --outfmt option.

```
VAT view -p 32 -a output.vatr -o output.tab -f tab
```

Tabular Format (-f tab)

VAT also supports a BLAST-like tabular output format, which is compact and easy to parse for downstream analysis.

Tabular Output Fields

- **Qseqid** – Query sequence identifier (read name).
- **Sseqid** – Subject sequence identifier (reference or database sequence name).
- **Pident** – Percentage identity within the aligned region, ranging from 0 to 100.
- **Length** – Total alignment length, including matches, mismatches, and gaps.
- **Mismatch** - Number of mismatching bases or amino acids in the alignment.
- **Gapopen** – Number of gap openings (insertion or deletion segments).
- **Qstart** – Alignment start position on the query sequence (1-based coordinate).
- **Qend** – Alignment end position on the query sequence (1-based coordinate).
- **Sstart** – Alignment start position on the subject (reference) sequence (1-based coordinate).
- **Send** – Alignment end position on the subject sequence (1-based coordinate).
- **Evalue** – Expectation value (E-value) of the alignment. Smaller values indicate more significant matches.
- **Bitscore** – Bit score of the alignment. Larger values indicate higher alignment quality.

Limitations

VAT requires hardware support for AVX2 (Advanced Vector Extensions 2). AVX2 is a CPU instruction set extension that enables high-performance vectorized operations. VAT uses these instructions to accelerate sequence alignment and improve overall computational efficiency. Because the alignment algorithms in VAT rely heavily on AVX2-based vectorization, the software requires CPUs that support this instruction set.

Checking AVX2 Support

You can verify whether your system supports AVX2 using the following commands.

On Linux:

Check for AVX2 support directly from CPU information:

```
grep -o 'avx2' /proc/cpuinfo | head -1
```

Alternatively, check the CPU flags using:

```
lscpu | grep -i avx2
```

If the output contains avx2, then your system supports AVX2 instructions.

Supported Processors

Most modern processors support AVX2, including:

- Intel CPUs: Haswell architecture and later
- AMD CPUs: Excavator architecture and later

Important Note

If the CPU does not support AVX2, VAT will not run. Users should ensure that their hardware platform supports AVX2 before compiling or executing VAT.